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| United States Patent | 12411514           |
| Kind Code            | B2                 |
| Date of Patent       | September 09, 2025 |
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### Trigger module with force feedback

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#### Abstract

A trigger module with force feedback includes a housing including an installation groove; a trigger swing bar rotatably connected to the housing; and a transmission device connected to the trigger swing bar. the transmission device includes a one-way transmission module engaged with the trigger swing bar, a driving module connected to the one-way transmission module and installed in the installation groove, and a driving member configured to drive the driving module. The driving member is connected to the driving module and installed in the housing; and the one-way transmission module includes two ends connected to the driving module and the trigger swing bar respectively. The trigger swing bar will not drive the driving member to reverse or drag the driving member when the trigger swing bar returns. Therefore, the trigger swing bar can quickly return, thereby providing users with good gaming experiences.

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| <b>Appl. No.:</b>            | <b>17/914782</b>                                                                                                          |
| <b>Filed (or PCT Filed):</b> | <b>April 26, 2022</b>                                                                                                     |
| <b>PCT No.:</b>              | <b>PCT/CN2022/089226</b>                                                                                                  |
| <b>PCT Pub. No.:</b>         | <b>WO2023/197376</b>                                                                                                      |
| <b>PCT Pub. Date:</b>        | <b>October 19, 2023</b>                                                                                                   |

Prior Publication Data

|                     |                  |
|---------------------|------------------|
| Document Identifier | Publication Date |
| US 20240393821 A1   | Nov. 28, 2024    |

Foreign Application Priority Data

|    |                |               |
|----|----------------|---------------|
| CN | 202220840600.9 | Apr. 12, 2022 |
|----|----------------|---------------|

Publication Classification

Int. Cl.: G05G1/04 (20060101); A63F13/24 (20140101); A63F13/28 (20140101); A63F13/285 (20140101); G05G5/03 (20080401)

U.S. Cl.:

CPC G05G1/04 (20130101); A63F13/24 (20140902); A63F13/285 (20140902); G05G5/03 (20130101); G05G2505/00 (20130101)

Field of Classification Search

CPC: G05G (1/04); G05G (5/03); G05G (5/05); G05G (9/047); G05G (2505/00); A63F (13/24); A63F (13/285)

References Cited

U.S. PATENT DOCUMENTS

| Patent No.   | Issued Date | Patentee Name | U.S. Cl. | CPC         |
|--------------|-------------|---------------|----------|-------------|
| 10773159     | 12/2019     | Schmitz       | N/A      | A63F 13/285 |
| 12079024     | 12/2023     | Li            | N/A      | G05G 5/03   |
| 12090392     | 12/2023     | Morita        | N/A      | A63F 13/285 |
| 12186655     | 12/2024     | Jiang         | N/A      | A63F 13/24  |
| 12222742     | 12/2024     | Li            | N/A      | G05G 5/03   |
| 2023/0400926 | 12/2022     | Warner        | N/A      | G06F 3/016  |
| 2024/0029971 | 12/2023     | Yokotsuka     | N/A      | H01H 21/36  |

FOREIGN PATENT DOCUMENTS

| Patent No.    | Application Date | Country | CPC         |
|---------------|------------------|---------|-------------|
| 113082690     | 12/2020          | CN      | A63F 13/218 |
| 113209605     | 12/2020          | CN      | A63F 13/24  |
| 214956567     | 12/2020          | CN      | N/A         |
| 113972086     | 12/2021          | CN      | N/A         |
| 215496419     | 12/2021          | CN      | N/A         |
| 215962111     | 12/2021          | CN      | N/A         |
| 218975323     | 12/2022          | CN      | N/A         |
| 220609067     | 12/2023          | CN      | N/A         |
| 118001714     | 12/2023          | CN      | N/A         |
| WO-2023244624 | 12/2022          | WO      | A63F 13/218 |

OTHER PUBLICATIONS

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## **Background/Summary**

### **TECHNICAL FIELD**

(1) The present invention relates to the technical field of electronic device, and in particular, to a trigger module with force feedback.

### **BACKGROUND**

(2) With the development of network technology and electronic information technology, more and more game development, various operation handles, virtual reality devices, and augmented reality devices are developed, and users have higher and higher sensory requirements when using peripheral devices. In related art, the triggers of various handles or peripheral devices cannot be adjusted according to different game scenarios, and some triggers only provide certain vibration feedback, which is difficult to provide users with good gaming experiences and cannot provide users with immersive gaming experiences.

### **SUMMARY**

(3) An embodiment of the present invention provides a trigger module with force feedback, including: a housing including an installation groove; a trigger swing bar rotatably connected to the housing; and a transmission device connected to the trigger swing bar. The transmission device includes a one-way transmission module engaged with the trigger swing bar, a driving module connected to the one-way transmission module, and a driving member configured to drive the driving module. The driving member is connected to the driving module and installed in the housing; the driving module is installed in the installation groove; and the one-way transmission module includes an end connected to the driving module, and another end engaged with the trigger swing bar.

(4) As an improvement, the one-way transmission module includes: a fixed part connected to an end of the driving module, a first elastic piece connected to the fixed part, at least one ratchet tooth connected to an end of the first elastic piece away from the fixed part, and a ratchet wheel detachably engaged with the at least one ratchet tooth. An upper part of the ratchet wheel is engaged with the trigger swing bar, an end of the at least one ratchet tooth away from the ratchet wheel is connected to the fixed part, and the first elastic piece is configured to provide the at least one ratchet tooth with an elastic force towards the ratchet wheel.

(5) As an improvement, the fixed part includes an accommodating groove for accommodating the at least one ratchet tooth, and the at least one ratchet tooth includes a body part connected into the accommodating groove and an engaging part arranged at an end of the body part, The engaging part is arranged in the accommodating groove.

(6) As an improvement, the ratchet wheel includes first gear teeth that are unidirectionally engaged with the at least one ratchet tooth, and a side of the engaging part is at least partially engaged with a corresponding first gear tooth of the first gear teeth.

(7) As an improvement, the first gear teeth include first engaging slopes spaced apart from each other about an axis of the ratchet wheel, and first guide surfaces each of which connects two adjacent first engaging slopes. The engaging part includes a second engaging slope matching the first engaging slope, and a second guide surface. The second guide surface includes an end

connected to the second engaging slope and another end connected to the body part. An inclination angle is formed between the first engaging slope and the first guide surface.

(8) As an improvement, the fixed part includes a limiting structure disposed in the accommodating groove. An end of the first elastic piece away from the at least one ratchet tooth is connected to the limiting structure.

(9) As an improvement, the upper part of the ratchet wheel includes a toothed structure engaged with the trigger swing bar.

(10) As an improvement, the trigger swing bar includes: a trigger detachably engaged with the one-way transmission module, a connecting shaft including an end mounted to the housing and another end connected to the trigger, and a second elastic piece sleeved on a periphery of the connecting shaft. The second elastic piece is arranged at the housing and configured to provide an elastic force to reset the trigger.

(11) As an improvement, the trigger includes: a connecting part engaged with the one-way transmission module, and a pressing part connected to the connecting part. A side of the connecting part connected to the one-way transmission module is provided with a gear structure engaged with the one-way transmission module.

(12) As an improvement, the driving module includes a worm wheel connected to the one-way transmission module and a worm connected to the worm wheel in a transmission way. The worm and the worm wheel are arranged in the installation groove, and the worm penetrates the housing and then is connected to an output shaft of the driving member.

(13) The beneficial effect of the present invention is as follows. A trigger module with force feedback is provided and includes the housing having the installation groove, the trigger swing bar rotatably connected to the housing, and the transmission device connected to the trigger swing bar; the transmission device includes the one-way transmission module engaged with the trigger swing bar, the driving module connected to the one-way transmission module, and the driving member configured to drive the driving module; the driving member is connected to the driving module and installed in the housing; the driving module is installed in the installation groove; and the one-way transmission module includes an end connected to the driving module, and another end engaged with the trigger swing bar. The force feedback trigger module provided by the present invention utilizes the driving member to drive the driving module to transmit the torque, and utilizes the one-way transmission module to unidirectionally transmit the torque to the trigger swing bar, that is, only a resistance for pressing the trigger swing bar is applied, and it can ensure that the trigger swing bar can be quickly reset after no pressing is applied on the trigger swing bar, the one-way transmission module can utilize the detachable engagement connection at the position where the one-way transmission module is engaged with the trigger swing bar and at the position where the one-way transmission module is connected to the driving module, to allow the trigger swing bar to be temporarily disengaged from the driving module when the trigger swing bar is reset. In this way, the trigger swing bar will not drive the driving member to reverse or drag the driving member when the trigger swing bar returns. Therefore, the trigger swing bar can quickly return, thereby providing users with good gaming experiences.

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## Description

### BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a schematic diagram of a trigger module with force feedback from a first angle provided by an embodiment of the present invention;
- (2) FIG. 2 is a schematic diagram of a trigger module with force feedback from a second angle provided by an embodiment of the present invention;
- (3) FIG. 3 is an exploded diagram of a trigger module with force feedback from a first angle

provided by an embodiment of the present invention;

(4) FIG. 4 is an exploded diagram of a trigger module with force feedback from a second angle provided by an embodiment of the present invention;

(5) FIG. 5 is a side view of a trigger module with force feedback provided by an embodiment of the present invention;

(6) FIG. 6 is a schematic plan view of a trigger module with force feedback from a first angle provided by an embodiment of the present invention;

(7) FIG. 7 is a schematic plan view of a trigger module with force feedback from a second angle provided by an embodiment of the present invention;

(8) FIG. 8 is a schematic plan view of a trigger module with force feedback from a third angle provided by an embodiment of the present invention; and

(9) FIG. 9 is a schematic plan view of a one-way transmission module provided by an embodiment of the present invention.

## DESCRIPTION OF EMBODIMENTS

(10) The present invention will be described below with reference to the accompanying drawings and embodiments.

(11) Referring to FIG. 1 and FIG. 2, an embodiment of the present invention provides a trigger module with force feedback, and the trigger module with force feedback includes a housing **1** having an installation groove **11**, a trigger swing bar **2** rotatably connected to the housing **1**, and a transmission device **3** connected to the trigger swing bar **2**. The transmission device **3** includes a one-way transmission module **31** engaged with the trigger swing bar **2**, a driving module **32** connected to the one-way transmission module **31**, and a driving member **33** configured to drive the driving module **32**. The driving member **33** is connected to the driving module **32** and installed to the housing **1**, the driving module **33** is installed in the installation groove **11**, and the one-way transmission module **31** includes one end connected to the driving module **32**, and another end engaged with the trigger swing bar **2**.

(12) In this embodiment, the driving member **33** is used to drive the driving module **32** to transmit a torque, and the one-way transmission module **31** is used to unidirectionally transmit the torque to the trigger swing bar **2**, that is, only a resistance for pressing the trigger swing bar **2** is applied, and it can ensure that the trigger swing bar **2** can be quickly reset after no pressing is applied, and the one-way transmission module **31** can utilize the detachable engagement connection at a position where the one-way transmission module **31** is engaged with the trigger swing bar **2** and at a position where the one-way transmission module **31** is connected to the driving module, to allow the trigger swing bar **2** to be temporarily disengaged from the driving module when the trigger swing bar is reset. In this way, the trigger swing bar **2** will not drive the driving member **33** to reverse or drag the driving member **33** when the trigger swing bar **2** returns. Therefore, the trigger swing bar **2** can quickly return, thereby providing users with good gaming experiences.

(13) Further, the driving module **32** includes a worm wheel **321** connected to the one-way transmission module **31**, and a worm **322** connected to the worm wheel **321** in a transmission manner. The worm wheel **321** and the worm **322** are installed in the installation groove **11**, and the worm **322** penetrates the housing **1** and then is connected to an output shaft of the driving member **33**.

(14) In an embodiment, the driving member **33** is a motor. In order to save space and provide a better layout so that an axis of the motor is located in a rotation plane of the trigger swing bar **2**, a transmission manner with the worm wheel and the worm is adopted, so that the transmission direction can be changed by 90 degrees, and at the same time, a larger reduction ratio can be obtained. The worm **322** is driven by the motor to rotate, the worm wheel **321** rotates with the worm **322**, and a torque is transmitted by the one-way transmission module **31** to the trigger swing bar **2**. The transmitted torque will produce a unidirectional pushing force on the trigger swing bar **2**, and the pushing force acts as the resistance when the finger presses the trigger swing bar **2**. By

controlling the motor, the magnitude of the torque, the action period, and the action time can be changed, so as to achieve an ideal force feedback effect. As an improvement, the motor in some embodiments adopts a direct current (DC) motor. Since the driving module **32** and the one-way transmission module **31** have a large transmission ratio, a DC motor with a high speed and a low torque can obtain a large torque after transmission. With the one-way transmission module **31**, the torque of the DC motor can be unidirectionally transmitted to the trigger swing bar **2**, that is, only a resistance for pressing the trigger swing bar **2** is applied, and it can ensure that the trigger swing bar **2** can be quickly reset after no pressing is applied. Since the one-way transmission module **31** can be temporarily disengaged from the trigger swing bar **2** when the trigger swing bar **2** is reset, the trigger swing bar **2** will not drive the motor to reverse or drag the motor when the trigger swing bar **2** returns. Therefore, the trigger swing bar **2** can quickly return. Due to the reduction ratio and the internal resistance of the motor, if the one-way transmission module **31** is not provided, when the trigger swing bar **2** is pressed and then is released, the reset process of the trigger swing bar **2** will be subject to the resistance of the driving module **32**, the one-way transmission module **31** and the motor, as a result, the reset process becomes slow, which affects the trigger swing bar **2**.

(15) Further, referring to FIG. **1** to FIG. **9**, the one-way transmission module **31** includes a fixed part **311** connected to an end of the driving module **32**, a first elastic piece **312** connected to the fixed part **311**, a ratchet tooth **313** connected to an end of the first elastic piece **312** away from the fixed part **311**, and a ratchet wheel **314** detachably engaged with the ratchet tooth **313**. An upper part of the ratchet wheel **314** is detachably engaged with the trigger swing bar **2**, and an end of the ratchet tooth **313** away from the ratchet wheel **314** is connected to the fixed part **311**. The first elastic piece **312** is configured to provide the ratchet tooth **313** with an elastic force towards the ratchet wheel **314**. The fixed part **311** has an accommodating groove **3111** for accommodating the ratchet tooth **313**, and the ratchet tooth **313** includes a body part **3131** connected into the accommodating groove **3111** and an engaging part **3132** provided at an end of the body part **3131**. The engaging part **3132** is arranged in the accommodating groove **3111**. The ratchet wheel **314** includes first gear teeth **3141** unidirectionally engaged with a connection side of the ratchet teeth **313**. A side of the engaging part **3132** is at least partially engaged with a corresponding first gear tooth **3141**. The first gear teeth **3141** include first engaging slopes **31411** spaced apart from each other about an axis of the ratchet wheel **314**, and first guide surfaces **31412** each of which connects two adjacent first engaging slopes **31411**. The engaging part **3132** has a second engaging slope **31321** matching the first engaging slope **31411**, and a second guide surface **31322**. The second guide surface **31322** includes an end connected to the second engaging slope **31321** and another end connected to the body part **3131**. An inclination angle is formed between the first engaging slope **31411** and the first guide surface **31412**.

(16) In some embodiments, the structure of the one-way transmission module **31** adopts the transmission mode of a single row of the ratchet teeth **313** and a single row of ratchet wheel **314**. The fixed part **311** and the worm wheel **321** are fixed to each other or formed into one piece. The first elastic piece **312** is fixed to the fixed part **311**, and another end of the first elastic piece **312** acts on the ratchet teeth **313**. The ratchet teeth **313** can make an end of the first elastic piece **312** rotate about the axis of the fixed portion **311**, and in a natural state, the first elastic piece **312** acts on the ratchet teeth **313** so that the ratchet teeth **313** are engaged with the first gear teeth **3141** on the ratchet wheel **314**. An end of the ratchet wheel **314** away from the ratchet tooth **313** is engaged with the trigger swing bar **2**. When the fixed part **311** rotates clockwise with the worm wheel **321**, the ratchet teeth **313** is engaged with the ratchet wheel **314**, and at this time, the ratchet wheel **314** rotates synchronously with the fixed part **311**, in this case, torque can be transmitted. When the fixed part **311** rotates counterclockwise, the ratchet teeth **313** on the fixed part **311** is engaged with the ratchet wheel **314** due to the tooth-shaped structure of the ratchet wheel **314**, and at this time, the fixed part **311** rotates and the ratchet wheel **314** does not rotate. Similarly, when the ratchet wheel **314** rotates clockwise, due to the tooth-shaped structure of the ratchet teeth **313** and the

tooth-shaped structure of the ratchet wheel **314**, the ratchet wheel **314** will not drive the fixed part **311** to rotate, so the transmission of the worm wheel **321** and the worm **322** and the motor will not be affected, thereby making the trigger swing bar **2** quickly return.

(17) The body part **3131** and the engaging part **3132** of the ratchet tooth **313** are arranged in the accommodating groove **3111**, so that the ratchet tooth **313** can rotate in the corresponding position, the body part **3131** is connected to the fixed part **311**, and the engaging part **3132** rotate in the accommodating groove **3111** to drive the ratchet wheel **314** to rotate in the engaging state. An inclination angle is formed between the first engaging slope **31411** and the first guide surface **31412**. When the fixed part **311** rotates clockwise with the worm wheel **321**, due to that a side of the engaging part **3132** is at least partially located in the first gear tooth **3141**, the first engaging surface abuts against and is engaged with the second engaging surface, and since the ratchet tooth **313** is fixed to the fixed part, the fixed part **311** can drive the ratchet wheel **314** to rotate synchronously at this time. When the fixed part **311** rotates counterclockwise, the first guide surface **31412** will be squeezed to the second guide surface **31322**, and the first elastic piece **312** is compressed, so the engaging part **3132** cannot be engaged with the first gear tooth **3141**, thereby ensuring that the fixed part **311** rotates and the ratchet wheel **314** does not rotate, or the ratchet wheel **314** will not drive the fixed part **311** to rotate, so the transmission of the worm wheel **321** and the worm **322** and the motor will not be affected, thereby making the trigger swing bar **2** quickly return.

(18) Further, referring to FIG. 3, the fixed part **311** has a limiting structure **3112** disposed in the accommodating groove **3111**, and an end of the first elastic piece **312** away from the ratchet teeth **313** is connected to the limiting structure **3112**.

(19) In the embodiments, the fixed part **311** is provided with a limiting structure **3112** corresponding to the first elastic piece **312**, so that the first elastic piece **312** is limited, and a force supplied on the ratchet tooth **313** towards the ratchet wheel **314** is continuously supplied. The force does not need to be too large, which may easily increase the frictional force relative to the ratchet wheel **314**. In some embodiments, the first elastic piece **312** adopts a torsion spring. There may be one or more evenly distributed combinations, and each combination includes the ratchet tooth **313** and the torsion spring. In some embodiments, there are four combinations each formed by the ratchet tooth **313** and the torsion spring, and the four combinations are arranged at equal angular intervals about the axis of the fixed part **311**. In other embodiments, the first elastic piece **312** can be a spring, and the number of the combinations each formed by the ratchet tooth **313** and the spring may be 2, 3, 5, or more.

(20) Further, referring to FIG. 6 and FIG. 9, the upper part of the ratchet wheel **314** has a toothed structure **3142** engaged with the trigger swing bar **2**.

(21) In some embodiments, the toothed structure **3142** of the ratchet wheel **314** at an end of ratchet wheel **314** away from the ratchet tooth **313** is a pinion, which can be engaged with the trigger swing bar **2**. When pressing the trigger swing bar **2**, the driving member **33** can rotate to supply a force, for resisting the pressing by the finger, on the trigger swing bar **2** through the driving module **32** and the one-way transmission module **31**. When the trigger swing bar **2** is released, a direction of the trigger swing bar **2** is rotated by the torsion spring, and in this case, the ratchet wheel **314** rotates but does not drive the fixed part **311** to rotate, that is, not drag the driving member **33**.

(22) Further, the trigger swing bar **2** includes a trigger **21** detachably engaged with the one-way transmission module **31**, a connecting shaft **22** including an end mounted to the housing **1** and another end connected to the trigger **21**, and a second elastic piece **23** sleeved on a periphery of the connecting shaft **22**. The second elastic piece **23** is arranged at the housing **1**, and the second elastic piece **23** is configured to provide an elastic force to reset the trigger **21**. The trigger **21** includes a connecting part **211** detachably engaged with the one-way transmission module, and a pressing part **212** connected to the connecting part **211**. A side of the connecting part connected to the one-way transmission module **31** has a gear structure **2111** detachably engaged with the one-way

transmission module.

(23) In the embodiments, the trigger **21** includes the connecting part **211** and the pressing part **212**, the pressing part **212** is a finger pressing part, and the connecting part **211** is a part having a toothed shape and engaged with the tooth structure **3142**. The housing **1** is provided with an installation hole **12**, the connecting shaft **22** passes through the connecting part **211** of the trigger **21** and is then rotatably connected to the installation hole **12**. The second elastic piece **23** is sleeved on a periphery of the installation hole **12**, and the second elastic piece **23** is arranged between the housing **1** and the connecting shaft **22** and is configured to provide a buffering force or rotational force. The connecting part **211** is provided with the gear structure **2111** matching the toothed structure **3142** and configured to detachably engage the connecting part **211** with the ratchet wheel **314**. In some embodiments, the second elastic piece **23** adopts a torsion spring. In other embodiments, the second elastic piece **23** can also be a spring.

(24) To sum up, the force feedback trigger module provided by the present invention utilizes the driving member to drive the driving module to transmit the torque, and utilizes the one-way transmission module to unidirectionally transmit the torque to the trigger swing bar, that is, only a resistance for pressing the trigger swing bar is applied, and it can ensure that the trigger swing bar can be quickly reset after no pressing is applied on the trigger swing bar, the one-way transmission module can utilize the detachable engagement connection at the position where the one-way transmission module is engaged with the trigger swing bar and at the position where the one-way transmission module is connected to the driving module, to allow the trigger swing bar to be temporarily disengaged from the driving module when the trigger swing bar is reset. In this way, the trigger swing bar will not drive the driving member to reverse or drag the driving member when the trigger swing bar returns. Therefore, the trigger swing bar can quickly return, thereby providing users with good gaming experiences.

(25) The above illustrates only some embodiments of the present invention. It should be noted that for those of ordinary skill in the art, improvements can be made without departing from the inventive concept of the present invention, and all these improvements shall fall within a scope of the present invention.

## Claims

1. A trigger module with force feedback, comprising: a housing comprising an installation groove; a trigger swing bar rotatably connected to the housing; and a transmission device connected to the trigger swing bar, wherein the transmission device comprises a one-way transmission module engaged with the trigger swing bar, a driving module connected to the one-way transmission module, and a driving member configured to drive the driving module, wherein the driving member is connected to the driving module and installed in the housing; the driving module is installed in the installation groove; and the one-way transmission module comprises an end connected to the driving module, and another end engaged with the trigger swing bar; wherein the one-way transmission module comprises a fixed part connected to an end of the driving module, a first elastic piece connected to the fixed part, at least one ratchet tooth connected to an end of the first elastic piece away from the fixed part, and a ratchet wheel detachably engaged with the at least one ratchet tooth; an upper part of the ratchet wheel is engaged with the trigger swing bar, an end of the at least one ratchet tooth away from the ratchet wheel is connected to the fixed part, and the first elastic piece is configured to provide the at least one ratchet tooth with an elastic force towards the ratchet wheel.

2. The trigger module with force feedback as described in claim 1, wherein the fixed part comprises an accommodating groove for accommodating the at least one ratchet tooth, and the at least one ratchet tooth comprises a body part connected into the accommodating groove and an engaging part arranged at an end of the body part, wherein the engaging part is arranged in the accommodating



groove.

3. The trigger module with force feedback as described in claim 2, wherein the ratchet wheel comprises first gear teeth that are unidirectionally engaged with the at least one ratchet tooth, and a side of the engaging part is at least partially engaged with a corresponding first gear tooth of the first gear teeth.
  4. The trigger module with force feedback as described in claim 3, wherein the first gear teeth comprise first engaging slopes spaced apart from each other about an axis of the ratchet wheel, and first guide surfaces each of which connects two adjacent first engaging slopes, wherein the engaging part comprises a second engaging slope matching the first engaging slope, and a second guide surface, wherein the second guide surface comprises an end connected to the second engaging slope and another end connected to the body part; and wherein an inclination angle is formed between the first engaging slope and the first guide surface.
  5. The trigger module with force feedback as described in claim 2, wherein the fixed part comprises a limiting structure disposed in the accommodating groove, wherein an end of the first elastic piece away from the at least one ratchet tooth is connected to the limiting structure.
  6. The trigger module with force feedback as described in claim 1, wherein the upper part of the ratchet wheel comprises a toothed structure engaged with the trigger swing bar.
  7. The trigger module with force feedback as described in claim 1, wherein the trigger swing bar comprises: a trigger detachably engaged with the one-way transmission module, a connecting shaft comprising an end mounted to the housing and another end connected to the trigger, and a second elastic piece sleeved on a periphery of the connecting shaft, wherein the second elastic piece is arranged at the housing and configured to provide an elastic force to reset the trigger.
  8. The trigger module with force feedback as described in claim 7, wherein the trigger comprises: a connecting part engaged with the one-way transmission module, and a pressing part connected to the connecting part, wherein a side of the connecting part connected to the one-way transmission module is provided with a gear structure engaged with the one-way transmission module.
  9. The trigger module with force feedback as described in claim 1, wherein the driving module comprises a worm wheel connected to the one-way transmission module and a worm connected to the worm wheel in a transmission way, wherein the worm and the worm wheel are arranged in the installation groove, and the worm penetrates the housing and then is connected to an output shaft of the driving member.
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